



Expressing an arithmetic sequence

1 In sequences, a position-to-term rule describes what? Tick **1** correct answer

- ☒ How to generate the term from the term number
- ☐ How many terms there are in a sequence
- ☐ How many positions there are in a sequence

2 Multiplying x by 4 and subtracting 7 is most simply written as which expression? Tick **1** correct answer

- ☐ $x \times 4 - 7$
- ☐ $x \times -7 + 4$
- ☐ $4 + x - 7$
- ☒ $4x - 7$

3 Which position-to-term rule satisfies this sequence? Tick **1** correct answer

Term number	1	2	3	4	5
Term value	1	4	7	10	13

- ☐ Add 3 each time.
- ☐ Starts at 1 and counts up in 3.
- ☐ Multiply by 3 and add 1
- ☒ Multiply by 3 and subtract 2



- 4 Which position-to-term rule links the pattern number to the number of squares? Tick **1** correct answer



- ☐ Multiply the pattern number by 4.
- ☒ Multiply the pattern number by 3 then add 1.
- ☐ Add 3 to the pattern number.
- ☐ Multiply the pattern number by 3 then subtract 1.

- 5 For the arithmetic sequence starting 1, 4, 7, 10, 13, ... the position-to-term rule is 'multiply by 3 and subtract 2'. What therefore, is the 100th term? Tick **1** correct answer

- ☐ 300
- ☐ 302
- ☒ 298
- ☐ 98

- 6 Substitute the value $a = 6$ into the below expressions. Order the value of the expressions from highest to lowest. Use numbers to show the correct order

1	$3a - 7$
4	$17 - 3a$
3	$7 - a$
2	$2a - 3$

